

FRC Research — Week 3 Working Notes

Framework v0.1, Smoke Test, and Baseline Reproduction

Juan Pablo Solís Ruiz
Supervisor: Prof. Ren Haijun

September 22, 2026

Contents

1	Purpose of These Notes	2
2	Handoff from Week 2	2
3	Week 3 Objective	3
4	Development Rules for This Phase	3
5	Framework v0.1 Scope	4
6	Initial Framework Structure	4
7	Configuration Design	5
7.1	Case Metadata	5
7.2	Physics	5
7.3	Geometry	6
7.4	Numerics	6
7.5	Runtime and HPC	6
8	Validation Philosophy	7
9	Run Provenance	8
10	Smoke Test	8
11	2000-Step Regression Run	9
12	After Regression: Scientific Formation Baseline	10
13	Week 3 Milestones	10
14	Current Research Log	11

1. Purpose of These Notes

These are living research notes for the third week of the FRC project.

Week 2 completed the takeover and characterization of the inherited WarpX-based formation workflow. The inherited 2000-step case is now sufficiently understood to serve as a quantitative regression reference.

The focus of Week 3 therefore changes from

understanding inherited work

to

building an independently controlled and reproducible FRC workflow

without changing the validated baseline physics.

The principal objective of this week is to construct the first minimal version of the FRC simulation framework, verify the complete execution pipeline with a small smoke test, and then reproduce the inherited 2000-step baseline using the new workflow.

2. Handoff from Week 2

The takeover audit established the following working interpretation of the inherited formation case:

- WarpX is used in 3D hybrid-PIC mode with MPI and OpenMP on CPU.
- The inherited reference simulation evolves for 2000 steps with

$$\Delta t = 1 \text{ ns},$$

corresponding to

$$t_{\text{end}} = 2 \text{ }\mu\text{s}.$$

- The imposed magnetic-field rise time is

$$t_{\text{peak}} = 4 \text{ }\mu\text{s},$$

so the inherited case terminates before completion of the full field ramp.

- Two reference realizations, i and j, use the same macroscopic physical and numerical inputs but different stochastic PIC particle realizations.
- The two runs are not bitwise identical, but their macroscopic magnetic and density evolution is highly reproducible.
- A coherent reversed-field structure develops during the available interval.
- Strong compression occurs near

$$t \approx 1.7 \text{ }\mu\text{s},$$

followed by partial magnetic relaxation by

$$t = 2 \text{ }\mu\text{s}.$$

The inherited case is therefore classified as a

computational regression baseline

rather than the final scientific formation simulation.

The longer scientific formation case will be defined only after the new framework reproduces this reference successfully.

3. Week 3 Objective

The main Week 3 objective is:

Build Framework v0.1, execute a minimal smoke test, and reproduce the inherited 2000-step formation baseline without changing its physical model.

The intended progression is

Framework v0.1 → Smoke Test → 2000-Step Regression → Scientific Formation Case
--

The first three stages belong to verification and software engineering.

The final stage begins the next physical study.

4. Development Rules for This Phase

The following rules are used to avoid mixing software refactoring with changes to the physical model.

1. The inherited reference remains read-only.

The directories

~/WQ/FRC ~/WQ/WarpX

are treated as legacy reference environments.

No `git reset`, `git clean`, `git pull`, `rebuild`, or direct editing should be performed there during baseline reproduction.

2. Framework development occurs independently.

The working area for this phase is

~/WQ/FRC_jpsr/frc_framework_v0_1

together with the previously created analysis tools under

~/WQ/FRC_jpsr/analysis

3. Physics and refactoring are not changed simultaneously.

The first full regression run must preserve the inherited physics and numerical model.

4. Generated WarpX inputs must remain readable.

The framework should organize and validate the model, not hide WarpX behind an opaque software layer.

5. Every new run must be self-describing.

Configuration, executable provenance, Slurm parameters, generated inputs, and output paths should be recoverable after the run finishes.

5. Framework v0.1 Scope

Framework v0.1 is deliberately minimal.

It is not intended to be a complete simulation platform yet.

Its first responsibility is to provide a controlled pipeline

```
structured case configuration
|
v
validation
|
v
generated WarpX input
|
v
generated Slurm job
|
v
WarpX
|
v
recorded outputs
|
v
run manifest
```

The framework should initially support only the inherited FRC formation workflow.

Tilt, Grad-Shafranov initialization, compression studies, parameter sweeps, and more advanced modules can be added after the formation regression baseline is reproduced.

6. Initial Framework Structure

The target structure for the first implementation is

```
frc_framework_v0_1/
|-- configs/
|   '-- formation/
|       |-- baseline.toml
|       '-- smoke.toml
|
|-- src/
|   '-- frc/
|       |-- __init__.py
|       |-- config.py
|       |-- validation.py
|       |-- input_generator.py
|       '-- manifest.py
|
|-- templates/
|   |-- warpx_formation.in
|   '-- scnet_cpu.sbatch
|
|-- scripts/
|   |-- generate_case.py
|   '-- submit_case.py
|
|-- runs/
|-- tests/
```

```
|-- docs/
|
'-- reference/
    '-- legacy_baseline/
        |-- input_test
        |-- run.sbatch
        |-- warpx_used_inputs_i
        '-- warpx_used_inputs_j
```

The `reference/legacy_baseline` files are snapshots used for comparison and should not be edited during framework development.

7. Configuration Design

The case configuration will become the single source of truth for each simulation.

The first version will use TOML because it is readable, explicit, and suitable for separating related parameter groups.

The configuration should distinguish at least the following categories.

7.1. Case Metadata

Examples:

- case name,
- description,
- parent/reference case,
- intended purpose such as smoke or regression.

7.2. Physics

Examples:

- initial density,
- electron temperature,
- ion temperature,
- bias magnetic field,
- driven magnetic field,
- field rise time,
- ion species and mass,
- resistive and hyper-resistive model parameters.

7.3. Geometry

Examples:

- chamber radius,
- physical axial length,
- simulation-domain dimensions,
- embedded-boundary parameters,
- PML configuration.

7.4. Numerics

Examples:

- grid resolution,
- timestep,
- total number of steps,
- particles per cell,
- particle shape,
- filters,
- hybrid-solver substeps,
- diagnostic cadence.

7.5. Runtime and HPC

Examples:

- WarpX executable,
- SCNet partition,
- number of nodes,
- MPI ranks,
- OpenMP threads,
- walltime,
- checkpoint behavior.

The long-term objective is that physical parameter studies can be performed by changing a case configuration rather than manually editing multiple scripts.

8. Validation Philosophy

Framework v0.1 should catch simple but scientifically important inconsistencies before a simulation is submitted.

The most immediate example comes directly from the inherited baseline.

The total physical runtime is

$$t_{\text{end}} = N_t \Delta t.$$

For the inherited case,

$$N_t = 2000, \quad \Delta t = 1 \text{ ns},$$

so

$$t_{\text{end}} = 2 \text{ } \mu\text{s}.$$

Since

$$t_{\text{peak}} = 4 \text{ } \mu\text{s},$$

the framework should automatically detect

$$t_{\text{end}} < t_{\text{peak}}$$

and produce a warning similar to

```
WARNING:
simulation end time = 2.0 us
field-ramp peak time = 4.0 us
This case terminates before completion of the imposed field ramp.
```

This warning should not prevent execution because the 2000-step case is a deliberate regression reference.

It should, however, prevent this condition from remaining scientifically implicit.

Additional first-version checks should include:

- positive grid dimensions,
- positive timestep,
- positive particle count,
- positive number of timesteps,
- diagnostic intervals compatible with the total run length,
- valid MPI/OpenMP resource counts,
- explicit output directory,
- detection of accidental case duplication where practical.

9. Run Provenance

Each independently executed case should create a run manifest.

A minimal manifest should eventually record:

- case name,
- creation time,
- source configuration,
- generated WarpX input,
- WarpX version and commit,
- WarpX executable checksum,
- framework version or Git commit,
- SCNet partition,
- Slurm job ID,
- node count,
- MPI ranks,
- OpenMP threads,
- walltime request,
- output directory,
- and reference/parent case.

The goal is that a future result can be traced from a figure or run directory back to the exact simulation configuration that produced it.

10. Smoke Test

The first independent execution under Framework v0.1 will be a smoke test.

The smoke test is **not** intended to validate FRC physics.

Its only purpose is to verify that the complete execution path works:

config → validation → WarpX input → Slurm → WarpX → output

The smoke case should preserve the essential model configuration while reducing computational and storage cost.

The initial target is approximately

10–20

timesteps, with minimal diagnostics and without the large full checkpoints used in the inherited production run.

A successful smoke test should demonstrate:

- the generated WarpX input is valid,
- the inherited WarpX executable can run from the new workspace,

- output paths are correctly isolated,
- the Slurm environment is correct,
- the manifest is created,
- and the resulting output can be read by the existing analysis tools.

11. 2000-Step Regression Run

After the smoke test succeeds, the inherited 2000-step case will be reproduced through the new framework.

This run is the principal verification milestone for Framework v0.1.

Because the particle initialization is stochastic, successful reproduction does not require byte-for-byte equality with either inherited reference run.

Instead, the new run should reproduce the established macroscopic behavior.

The principal regression observables are

$$B_{z,\text{center}}(t),$$

$$B_{z,\text{min}}(t),$$

$$B_{z,\text{max}}(t),$$

$$\rho_{\text{center}}(t),$$

and

$$\rho_{\text{max}}(t).$$

Spatial diagnostics should also reproduce the qualitative development of the reversed-field region.

The expected reference sequence is

initial bias $\rightarrow$ field reversal $\rightarrow$ strong compression $\rightarrow$ partial relaxation

with the strongest compression occurring near

$$t \approx 1.7 \mu\text{s}.$$

Pointwise quantities such as

$$T_{i,\text{max}}$$

should not initially be used as strict pass/fail criteria because they show substantially greater sensitivity to local PIC noise.

12. After Regression: Scientific Formation Baseline

Only after the 2000-step regression succeeds should the physical problem be extended.

The first scientific formation case should evolve through at least

$$t_{\text{peak}} = 4 \mu\text{S},$$

and probably beyond this time.

This longer case will be used to determine what happens after completion of the external field ramp and to begin quantitative analysis of

- magnetic topology,
- flux retention,
- density and temperature evolution,
- particle losses,
- energy evolution,
- FRC lifetime,
- and later stability.

At that stage, reconstruction of the poloidal flux function

$$\psi(r, z)$$

should be developed in order to identify the true separatrix and derive quantities such as

$$R_s, \quad L_s, \quad E = \frac{L_s}{2R_s}.$$

13. Week 3 Milestones

The intended Week 3 milestones are:

- ☐ Create the Framework v0.1 directory structure.
- ☐ Freeze the minimal inherited reference files required for regression.
- ☐ Define `baseline.toml` as the structured representation of the inherited 2000-step case.
- ☐ Derive `smoke.toml` from the baseline while preserving the physical model.
- ☐ Implement configuration loading.
- ☐ Implement first validation rules.
- ☐ Implement readable WarpX input generation.
- ☐ Implement Slurm job generation/submission.
- ☐ Implement run-manifest creation.
- ☐ Execute the first smoke test.
- ☐ Inspect and validate smoke-test output.

- ☐ Execute the independent 2000-step regression run.
- ☐ Compare the new run against inherited references `i` and `j`.
- ☐ Document any differences before introducing optimization or new physics.

14. Current Research Log

2026-09-22 — Start of Week 3

The Week 2 takeover audit is considered complete.

The inherited 2000-step FRC formation case has been characterized both computationally and physically and can now serve as the first regression reference.

The project focus has shifted to Framework v0.1.

The immediate next task is to create and verify the framework skeleton and then encode the inherited formation case in

```
configs/formation/baseline.toml
```

before implementing the corresponding smoke-test case.

Next entry

Continue with:

- Framework v0.1 skeleton verification;
- `baseline.toml`;
- `smoke.toml`;
- configuration loading and validation.

15. Research Log — 2026-09-22

The main objective of today's session was to begin the implementation of Framework v0.1 using the inherited 2000-step formation case as the regression reference.

The framework directory structure was created under

```
~/WQ/FRC_jpsr/frc_framework_v0_1
```

with separate locations for configurations, source code, templates, scripts, runs, tests, documentation, and frozen legacy-reference files.

The following inherited files were copied into the framework reference area:

```
reference/legacy_baseline/input_test
reference/legacy_baseline/run.sbatch
reference/legacy_baseline/warpx_used_inputs_i
reference/legacy_baseline/warpx_used_inputs_j
```

These files are treated as read-only reference material.

15.1. Baseline Configuration

The inherited formation case was encoded in

```
configs/formation/baseline.toml
```

The configuration separates the case into logical sections describing

- geometry,
- PML configuration,
- plasma parameters,
- magnetic drive,
- numerical parameters,
- boundary conditions,
- hybrid-PIC model parameters,
- diagnostics,
- checkpoints,
- SCNet runtime configuration,
- reference metadata,
- and regression metadata.

The first pasted version of the TOML file contained several truncated lines. Rather than repairing the file manually section by section, it was replaced with a clean copy and independently checked with a TOML parser before continuing.

The final configuration passed the syntax check successfully.

15.2. Configuration Loader and Validation Layer

The first software components of Framework v0.1 were implemented:

```
src/frc/config.py
src/frc/validation.py
scripts/validate_case.py
```

The configuration loader reads the TOML case and derives quantities that do not need to be stored explicitly in the user configuration.

For the inherited baseline, the framework derived

$$L_x = L_y = 0.7392 \text{ m},$$

$$\Delta x = \Delta y = 0.01155 \text{ m},$$

$$\Delta z = 0.0133333 \text{ m},$$

and the total axial grid including the in-domain PML as

$$N_{z,\text{total}} = 208.$$

The physical runtime was correctly reconstructed as

$$t_{\text{end}} = N_t \Delta t = 2000 \times 10^{-9} \text{ s} = 2 \text{ } \mu\text{s}.$$

The imposed magnetic-field rise time is

$$t_{\text{peak}} = 4 \text{ } \mu\text{s}.$$

Therefore, the framework correctly detected that the inherited regression case terminates before completion of the imposed field ramp.

At the final time of the regression case, the temporal drive factor is

$$\sin\left(\frac{\pi t_{\text{end}}}{2t_{\text{peak}}}\right) = \sin\left(\frac{\pi}{4}\right) \simeq 0.7071.$$

This condition is expected for the inherited regression case and is therefore reported as a warning rather than an error.

15.3. First Successful Framework Validation

The command

```
python scripts/validate_case.py \
    configs/formation/baseline.toml
```

completed successfully with

```
VALIDATION PASSED (1 warning(s))
```

The validator also confirmed

- the physical grid dimensions,
- the 208-cell axial domain including PML,
- the 32-thread execution layout,
- the inherited WarpX executable,
- and the presence of all four frozen legacy-reference files.

The intended SCNet execution layout was reconstructed as

1 node, 4 MPI ranks, 8 OpenMP threads/rank,

for a total of

32

CPU threads.

15.4. Status at End of Session

The first stage of Framework v0.1 is now working:

baseline.toml → configuration loader → derived quantities → validation

No WarpX job has yet been submitted from the new framework.

The next development step will be to implement

`src/frc/input_generator.py`

so that the structured TOML configuration can generate a human-readable WarpX input.

The generated input will first be compared against the frozen inherited `input_test` before any smoke-test execution is attempted.

The immediate sequence for the next session is therefore

generate WarpX input → compare with legacy input → define smoke case → run smoke test